James Rickards

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https://jamesrickards-canada.github.io/
https://github.com/JamesRickards-Canada

McNally North 120, Department of Mathematics and Computing Science
Saint Mary's University
Halifax, NS

Positions

Assistant Professor 2024 - present
Saint Mary's University Halifax, NS

Postdoctoral Fellow | ***Mentor: Katherine E. Stange*** 2021 - 2024
University of Colorado Boulder Boulder, CO

Education

Doctor of Philosophy | ***Advisor: Henri Darmon*** 2016 - 2021
McGill University Montreal, QC
Thesis title: Intersections of closed geodesics on Shimura curves

Master of Arts 2019
Trinity College, University of Cambridge Cambridge, UK

Master of Mathematics 2015 - 2016
Trinity College, University of Cambridge Cambridge, UK

Bachelor of Arts (Hons) | ***Major: Mathematics*** 2012 - 2015
Trinity College, University of Cambridge Cambridge, UK

Research Interests

Computational number theory, algebraic number theory, thin (semi)groups, arithmetic Fuchsian/Kleinian groups, binary quadratic forms, quaternion algebras, Shimura curves, circle packings, continued fractions, visualization.

Publications and Preprints

11. Eisenstein circle packings and the Eisenpint Schmidt arrangement 2026
James Rickards, Katherine E. Stange
Preprint

10. Primes represented by shifted quadratic forms: on primitivity and congruence classes 2026
Elena Fuchs, Catherine Hsu, James Rickards, Damaris Schindler, Katherine E. Stange
Acta Arithmetica 222 (2026), 371-391

9. Prime and thickened prime components in Apollonian circle packings 2024
Elena Fuchs, Holley Friedlander, Piper Harris, Catherine Hsu, James Rickards, Katherine Sanden, Damaris Schindler, Katherine E. Stange
Accepted to Proceedings of Women in Number Theory VI.

8. Reciprocity obstructions in semigroup orbits in SL$(2,\mathbb{Z})$ 2025
James Rickards, Katherine E. Stange
Duke Mathematical Journal 174(15): 3197-3244 (15 October 2025)

7. The local-global conjecture for Apollonian circle packings is false 2024
Summer Haag, Clyde Kertzer, James Rickards, Katherine E. Stange
Annals of Mathematics (2) 200(2): 749-770 (September 2024)

6. The Apollonian staircase 2024
James Rickards
International Mathematics Research Notices, Volume 2024, Issue 2, January 2024, pp. 1340-1372

5. Improved computation of fundamental domains for arithmetic Fuchsian groups 2022
James Rickards
Mathematics of Computation **91** (2022), no. 338, pp. 2929-2954

4. Hecke operators acting on optimal embeddings in indefinite quaternion algebras 2022
James Rickards
Acta Arithmetica **204** (2022) no. 4, pp. 347-367

3. Counting intersection numbers of closed geodesics on Shimura curves 2023
James Rickards
Research in Number Theory **9** (2023), no. 2, Paper No. 20, 45 pp.

2. Computing intersections of closed geodesics on the modular curve 2021
James Rickards
Journal of Number Theory, **225** (2021), pp. 374-408

1. When is a Polynomial a Composition of Other Polynomials? 2011
James Rickards
American Mathematical Monthly, **118** (2011), no. 4, pp. 358-363

Research Grants

Total funding for each grant is listed. All amounts are in CAD.

NSERC Discovery Grant 2025 - 2030
Thin groups, quaternion algebras, and computational number theory $187,500

FGSR Supplementary Funding for International Conference Participation for Faculty Members 2024
Arithmetic groups, hyperbolic manifolds and computation $1,000

FGSR University Grant in Aid of Research 2024 - 2025
Computational number theory $5,000

SMU Start-up Funding Package 2024 - 2027
$26,500

Media

CU students follow their noses, disprove math conjecture 2023
Article about *The Local-Global Conjecture for Apollonian circle packings is false*
Colorado Arts and Sciences Magazine,
https://www.colorado.edu/asmagazine/2023/11/30/cu-students-follow-their-noses-disprove-math-conjecture

The Hidden Connection That Changed Number Theory 2023
Contributed quotes
Quanta Magazine, https://www.quantamagazine.org/the-hidden-connection-that-changed-number-theory-20231101/

Two Students Unravel a Widely Believed Math Conjecture 2023
Article about *The Local-Global Conjecture for Apollonian circle packings is false*
Quanta Magazine, https://www.quantamagazine.org/two-students-unravel-a-widely-believed-math-conjecture-20230810/

Code

Apollonian PARI/GP
Computations for Apollonian circle packings, including basic operations, generating pictures in LaTeX, and a very efficient implementation for finding all missing curvatures up to a bound.
Available at https://github.com/JamesRickards-Canada/Apollonian

Apollonian-Prime PARI/GP
Computations for thickened prime components of Apollonian circle packings,
Available at https://github.com/JamesRickards-Canada/Apollonian-Prime

Eisenstein PARI/GP
Computations for Eisenstein circle packings, including basic operations, generating pictures in LaTeX, generating circle in the full Eisenstein/Eisenpint Schmidt arrangements, and computing all curvatures in a packing.
Available at https://github.com/JamesRickards-Canada/Eisenstein

Fundamental domains for Shimura curves PARI/GP

Computation of fundamental domains for arithmetic Fuchsian groups. Improves on the algorithms of Voight and Page, and is significantly more efficient than the live Magma implementation (from 100 to millions of times as fast, depending on the example). Will be integrated into PARI/GP.

Available at https://github.com/JamesRickards-Canada/Fundamental-Domains-for-Shimura-curves

Isogeny PARI/GP, Sage

Computation of supersingular ℓ and L isogeny graphs, significantly more efficient than the live Sage implementation. Includes code to seamlessly use it inside of Sage.

Available at https://github.com/JamesRickards-Canada/Isogeny

Q-Quadratic PARI/GP

Computing with integral binary quadratic forms and quaternion algebras over $\mathbb{Q}$. Includes algorithms to compute intersection numbers of modular geodesics, as described in my thesis and various papers.

Available at https://github.com/JamesRickards-Canada/Q-Quadratic

Semigroup Reciprocity PARI/GP

Computation of orbits of semigroups, including efficient implementation of missing numbers in an orbit. This package accompanies the paper *Reciprocity obstructions in semigroup orbits in* $SL(2, \mathbb{Z})$, and includes methods to check various results.

Available at https://github.com/JamesRickards-Canada/Semigroup-Reciprocity

Other academic writing

Competition Highlights: Canadian Mathematical Olympiad and Junior Olympiad (CMO/CJMO)

Paweł Prałat, James Rickards

Crux Mathematicorum: Vol. 50(8), October 2024; Vol. 51(7), September 2025

Running PARI/GP on Windows

Tutorial for installing and using PARI/GP on Windows computers.

Available at https://pari.math.u-bordeaux.fr/PDF/PARIGP-Windows.pdf

Polynomial Division in Number Theory

Crux Mathematicorum, Vol. 43(10), December 2017

Parametric Solutions to the Generalized Fermat Equation

Part III essay, Cambridge, 2016

Higher Power Reciprocity Laws

Rouse Ball Mathematical Essay, Cambridge, 2015

Conference Talks

Atelier libpari 2025 Jun 2025

Supersingular Isogeny Graphs (in PARI/GP) Université de Bordeaux

Arithmetic groups, hyperbolic manifolds and computation Dec 2024

Totally geodesic surfaces in Bianchi orbifolds Université de Bordeaux

ANTS XVI Jul 2024

Reciprocity obstructions in continued fraction semigroups MIT

Computational Aspects of Thin Groups Jun 2024

The not-so-local-global conjecture NUS

Renormalization, computation and visualization in Geometry, Number Theory and Dynamics Sept 2023

The not-so-local-global conjecture CIRM

LuCaNT Jul 2023

Software demo: Computing fundamental domains for congruence arithmetic Fuchsian groups in PARI/GP ICERM

Number Theory Informed by Computation Aug 2022

Fast fundamental domains for arithmetic Fuchsian groups in PARI/GP Park City Mathematics Institute

16th Atelier PARI/GP 2022 Jan 2022

Fundamental Domains for Shimura curves U. Franche-Comté (participated online)

Lattices and Cohomology of Arithmetic Groups: Geometric and Computational Viewpoints Oct 2021
Improved computation of fundamental domains for arithmetic Fuchsian groups BIRS (online)

Front Range Number Theory Day Sep 2021
Counting intersection numbers on Shimura curves Colorado State University

Front Range Number Theory Day Apr 2021
Fast computations of fundamental domains for Shimura curves CU Boulder (online)

Quebec-Maine Number Theory Conference Sep 2020
Computing with (indefinite) quadratic forms and quaternion algebras in PARI/GP Laval University (online)

Quebec-Maine Number Theory Conference Oct 2019
Intersection numbers of modular geodesics University of Maine

Quebec-Maine Number Theory Conference Oct 2018
Intersection numbers of modular geodesics Laval University

CMS Summer Meeting Jun 2018
Number theoretic intersection numbers on Riemann surfaces University of New Brunswick

Montreal-Toronto Workshop in Number Theory Dec 2016
Basic background on mock modular forms and weak harmonic Maass forms University of Montreal

Seminar Talks

The Ohio State University Number Theory Seminar Apr 2025
Failure of the local-global conjecture in thin (semi)groups The Ohio State University

Undergraduate Research Seminar Oct 2024
Apollonian circle packings and the not-so-local-global conjecture Saint Mary's University

Algebraic Geometry Seminar May 2024
The not-so-local-global conjecture UC Davis

PU/IAS Number Theory Seminar Apr 2024
The not-so-local-global conjecture Princeton University / IAS

Dalhousie Number Theory Seminar Mar 2024
Quaternion algebras in number theory Dalhousie University

Dalhousie Colloquium Mar 2024
The not-so-local-global conjecture Dalhousie University

Saint Mary's Colloquium Jan 2024
Apollonian circle packings and thin groups Saint Mary's University

Virtual Seminar on Geometry and Topology Nov 2023
Failure of the local-global conjecture in thin (semi)groups KIAS, South Korea

Penn State Algebra and Number Theory Seminar Oct 2023
The not-so-local-global conjecture Penn State

University of Washington Number Theory Seminar Oct 2023
The not-so-local-global conjecture University of Washington

Arithmetic Reflection Groups Seminar Aug 2023
The not-so-local-global conjecture Online

Five College Number Theory Seminar Nov 2022
The Apollonian Staircase Amherst College

Brown University Algebra and Algebraic Geometry Seminars Nov 2022
The Apollonian Staircase Brown University

International Seminar on Automorphic Forms May 2021
Counting intersection numbers on Shimura curves TU Darmstadt/ETH Zurich (online)

Rutgers Number Theory Seminar Oct 2019
Intersection numbers of modular geodesics Rutgers University

Laval Number Theory Seminar Oct 2019
Intersection numbers of modular geodesics Laval University

Teaching Experience - Saint Mary's University

Math 2305 | ***Survey of Discrete Mathematics*** F24 + 2R, F25 + 1R, F26 + 2R

Math 2310 | ***Introductory Analysis*** W25 + 1R

Math 3025* | ***Introduction to Number Theory*** W25, F26

Math 3441 | ***Real Analysis I*** F25

Math 4420 | ***Abstract Algebra I*** W26

Math 4426 | ***Theory of Functions of a Complex Variable I*** W27

* Math 3025 was taught once as Math 3827 (Special Topic: Number Theory) before being added as a permanent course.

Teaching Experience - University of Colorado Boulder

Math 2001 | ***Introduction to Discrete Mathematics*** Fall 2022 - 2 sections, Spring 2024

Math 2130 | ***Linear Algebra for Non-Math Majors*** Fall 2021, Spring 2022

Math 3001 | ***Analysis 1*** Fall 2023

Math 3110 | ***Introduction to the Theory of Numbers*** Spring 2022, Spring 2024

Math 8174 | ***Topics in Algebra - Quaternion Algebras (Graduate course)*** Spring 2023

Teaching Experience - other

TA for PCMI graduate course Summer 2022
TA for Jan Vonk's one week long course at the Park City Mathematics Institute graduate summer school

Math 141 TA | ***Integral Calculus*** Fall 2017, Fall 2018
McGill University

Mentorship

Summer Research - Saint Mary's
Advised summer undergraduate research students for 12-16 weeks on projects involving Apollonian circle packings. In 2025, advised 3 students. In 2026, advised 3 students.

Honours Thesis Advisor
Co-advisor to Clyde Kertzer on symmetries in Apollonian circle packings (Fall 2023 - Spring 2024).

2023 REU - CU Boulder
Ran an REU jointly with Katherine E. Stange on Apollonian circle packings. Supervised one undergraduate student (Clyde Kertzer) and one first year graduate student (Summer Haag).

Math camp leader and trainer 2015, 2017 - 2019
Mentored and trained Canadian high school students interested in contest math at four (week-long) IMO (International Mathematical Olympiad) winter camps, as well as four IMO summer camps (3 weeks long each), and one EGMO (European Girls Mathematical Olympiad) training camp (weekend).

Scholarships

Vanier Canada Graduate Scholarship 2018 - 2021
$\$50,000$ CAD/year

NSERC CGS D 2018 (Declined)

Schulich Fellowship | ***McGill University*** 2016 - 2018
$\$25,000$ CAD/year

Trinity College Woods Scholarship 2015 - 2016
$\$25,000$ CAD/year

Cambridge Trusts Scholarship 2015 - 2016
$\$25,000$ CAD/year

Blyth Cambridge Commonwealth Scholarship 2012 - 2015
$\$50,000$ CAD/year

Lazaridis Olympiad Scholarship to University of Waterloo 2012 (Declined)

Math Huskies

Science Atlantic Fall 2024 - present
Organize participation of Saint Mary's team in the Science Atlantic mathematics contest.

Weekly training sessions Fall 2024 - present
Run weekly training sessions during Fall semester preparing Saint Mary's University students for the Science Atlantic and Putnam math contests.

Canadian Mathematical Society Service

Canadian International Mathematical Olympiad Committee chair 2019 - present
Organize a 1 week training camp every January for 15-20 students, coordinate the writing and grading of the APMO contest for around 30 students, organize the team selection test and grading for 12-18 students, help plan the 2 week summer training camp and the IMO itself, plus general administrative duties.

Canadian Junior Mathematical Olympiad coordinator 2019 - present
This is a variant of the CMO for students in grade ≤ 10. I am in charge of ensuring that problems get selected for the contest.

Canadian International Mathematical Olympiad Committee member 2016 - present
Propose problems and participate in grading. This is a high school olympiad style contest for 70-100 Canadian students.

Canadian Open Mathematics Challenge Problems Committee member 2013 - 2021
Proposed problems and participated in the selection of the final contest.

International Mathematical Olympiad Service

Team Canada Leader Observer 2019
Help run the 1 week long winter training camp and 2 week long summer training camp, then head to the IMO early to participate in problem selection and grading of our student's problems.

Team Canada Leader 2017, 2018
The leader and leader observer have similar duties, but the leader is officially in charge.

Team Canada Deputy Leader Observer 2015
Similar duties to the leader, except the deputy leader and observer accompany the students to the IMO. They still participate in grading/coordination of the Canadian student submissions.

Other Mathematical Olympiad Service

Olympiade Francophone de Mathématiques 2021 - present
Organizer for the Canadian team

Paper review

Reviewed papers/project proposals for (among others): Acta Arithmetica, Banff International Research Station, Commentarii Mathematici Helvetici, Communications in Algebra, International Mathematical Research Notices, Journal of Number Theory, Journal of the European Mathematical Society, and Transactions of the American Mathematical Society.

Other service

Undergraduate thesis reviewer 2026 - present
Served as the reader for one undergraduate thesis. Saint Mary's

Code Huskies 2025 - present
Gave a couple of training sessions each semester for the SMU Code Huskies, an undergraduate programming contest club. Saint Mary's

Comprehensive exams 2021 - 2024
Committee member for three comprehensive oral exams at CU Boulder. CU Boulder

Skills

Languages: English (native), French (limited working proficiency)

Programming:

- High proficiency: C, PARI/GP
- Medium proficiency: Python
- Some familiarity: HTML, Magma, Mathematica, SageMath